Title

Impact of oral cholera vaccine outbreak response immunization in sub-Saharan Africa, 2010–2020: a modeling analysis

Abstract

Background

Cholera outbreaks remain a major public health threat in sub-Saharan Africa. The impact of oral cholera vaccines (OCVs) depends on deployment conditions. We evaluated how the health and economic impact of single-dose outbreak response immunization (ORIs) vary with outbreak characteristics and implementation parameters.

Methods

Using data from 1,192 cholera outbreaks (2010–2020) in sub-Saharan Africa, we simulated counterfactual ORIs varying by timing, coverage, triggering criteria, and outbreak selection strategy. Outcomes—evaluated at the individual-outbreak level and aggregated across subsets—included *impact* (cases, deaths, and DALYs averted), *efficiency* (impact per 1,000 OCV doses), and *cost-effectiveness* (cost per unit of impact). Stochastic simulations incorporated age-specific vaccine efficacy, indirect protection, delays in ORI initiation, and the time required to develop vaccine-derived immunity.

Findings

At 75% coverage, an ORI implemented in week 1 averted a median of 129 cases [IQR: 41–375; 70% of total]. Impact declined exponentially with implementation delay—by 23·6% per week , or halving every 2·6 weeks—falling to 16 cases (7·8%) by week 9, comparable to historical response times. Early ORIs averted 0·69 cases [IQR: 0·19-2·40] per 1,000 doses at a median cost of US$4,662 [IQR: 985-17,917] per case averted with the medians decreasing to 0·09 cases per 1,000 doses and increasing to US$13,043 per case averted by week 9. Prioritizing outbreaks in the top quartile of attack rate increased median case averted per 1000 OCVs by 5-15 fold and reduced the cost per DALY averted by 8-85 fold. Aggregated across outbreaks, reflecting OCV stockpile use, week-9 ORIs achieved 0.44 [IQR: 0.28 - 0.44] cases averted per 1,000 doses, rising to 2.2 [IQR: 1.9-2.2] when implemented in week 1 among the top 70% of outbreaks by attack rate.

Conclusion

Rapid OCV deployment and prioritization of large, sustained, high–attack-rate outbreaks are essential to maximize impact, efficiency and cost effectiveness. Even with delays, stockpile-supported campaigns can substantially reduce cholera burden in Africa.

Key messages

1. We analyzed 1,192 cholera outbreaks across sub-Saharan Africa (2010–2020) to estimate the impact, efficiency, and cost effectiveness of single-dose oral cholera vaccine (OCV) outbreak response immunization (ORIs).
2. ORI implemented in week 1 of outbreak averted a median 129 cases [IQR: 41–375] averted, corresponding to 70% [QR: 49.8 - 82.5]% reduction; however, impact declined sharply, halving each median 2.6 [IQR: 0.56.1] weeks of delay.
3. Prioritizing large, prolonged, and high-attack-rate outbreaks markedly improved impact, efficiency, and cost-effectiveness, with up to ~85-fold gains in cost-effectiveness among the top 10% of outbreaks by attack rate.
4. In simulations reflecting historical OCV stockpile use, even week-9 ORIs averted 0·44 [IQR: 0·28–0·44] cases per 1,000 doses, increasing to 2·2 [IQR:1·9–2·2] with week-1 campaigns among the top 70% of outbreaks by attack rate.
5. Rapid and targeted ORI deployment is essential to maximize the impact of limited global OCV supply and to reduce cholera burden in Africa.

Research in context

Evidence before this study

We searched PubMed on March 15, 2024, without language or date restrictions, using the search terms (("Cholera"[Mesh] OR "Vibrio cholerae"[Mesh])) AND "Cholera Vaccines"[Mesh] AND ((model*[Title/Abstract]) OR (math*[Title/Abstract])) AND (vaccine*[Title/Abstract]). The search yielded 164 articles, of which 17 studies met the inclusion criteria, focusing on mathematical models assessing the impact of oral cholera vaccines (OCVs) under different scenarios. Most studies simulated outbreaks in single geographic settings—such as Bangladesh[1–4], Chad[5], Haiti[6–8], South Sudan,[9] or Zimbabwe[10]—or not anchored in a specific setting.[11] A few analyses considered outbreaks across multiple countries.[12–16] These studies varied widely in campaign design (reactive mass campaigns, proactive campaigns, "hotspot" or area-targeted campaigns), outbreak scale, and modeling assumptions, including delays in vaccine delivery and onset of protection. While some models assumed protection immediately following vaccination[13,14,17,18], others considered a lag between vaccination and protection.[5,7,12,19] Most previous work focused on large, well-documented outbreaks (e.g., Zimbabwe with 98,591 cases or Haiti with 119,902 cases)[14,19] and did not represent the outbreaks that may be typically found in sub-Saharan Africa. These gaps highlight the need for models that capture heterogeneity of outbreaks to improve the generalizability of vaccine impact assessments.

Added value of this study

To our knowledge, this is the first study to model the impact, efficiency, and cost-effectiveness of OCV outbreak response immunizations (ORIs) across a large and diverse set of outbreaks from multiple temporal and geographic contexts in sub-Saharan Africa. Using data from 1,192 outbreaks from 2010 to 2020—spanning wide variation in size, duration, and attack rate—we developed a comprehensive framework to evaluate how vaccination timing, coverage, and targeting influence outcomes. Unlike previous studies focused on isolated large outbreaks, our model quantifies the distribution of vaccine impact across heterogeneous epidemics, providing a generalizable basis for reactive vaccination planning and optimization of global OCV stockpile use.

Implications of the available evidence

Our findings underscore the critical importance of rapid and targeted vaccine deployment. ORI impact declined by half approximately every 2·6 weeks of implementation delay, yet substantial benefits persisted even with late campaigns, particularly in large and high–attack-rate outbreaks. When aggregated across outbreaks, week-9 ORIs, comparable to historical response times, still averted 0·44 [IQR: 0·28–0·44] cases per 1,000 OCV doses, rising to 2·2 [IQR: 1·9–2·2] with week-1 campaigns targeting the top 70% of outbreaks by attack rate. Prioritizing large, sustained outbreaks increased cost-effectiveness of week-9 ORI by up to 85-fold. These results provide a quantitative basis for optimizing reactive vaccination strategies under constrained OCV supply to maximize health impact across Africa.

Main text

Introduction

Cholera, one of the oldest infectious diseases affecting humans, has persisted as a major public health challenge since the 7th pandemic began near the Bay of Bengal in 1961.[20] It remains a significant threat in settings with inadequate water, sanitation, and hygiene (WaSH), particularly in low- and middle-income countries (LMICs).[21] Modelled estimates and World Health Organization (WHO) reports indicate that sub-Saharan Africa bears most of the global burden, marked by substantial spatial heterogeneity.[22–24] An analysis of cholera incidence in the region from 2010 to 2016 identified nearly 1,000 outbreaks of varying scale across multiple administrative levels,[25] underscoring the complexity of controlling transmission.

Sustained access to adequate WaSH—together with case management and surveillance—remains the foundation of cholera control. However, oral cholera vaccines (OCVs) provide a critical complementary tool, particularly where long-term infrastructure improvements are difficult to achieve. To facilitate rapid vaccine access, WHO established the global OCV stockpile in 2013 with Gavi's support.[26] Between 2013 and 2024, more than 200 million doses have been deployed for both proactive and reactive campaigns.[27] In response to growing demand and limited supply, the International Coordinating Group (ICG)—which manages the stockpile—temporarily adopted a single-dose regimen for outbreak response in October 2022, replacing the standard two-dose schedule.[26]

Given constrained vaccine supply, optimizing the use of available doses is critical. While preemptive vaccination—targeting high-risk areas identified through environmental or historical risk indicators[28], or enhanced diagnostics[16]—can be effective, such approaches are challenged by fluctuating incidence,[29] waning immunity,[30] and population mobility.[17,31] In contrast, outbreak response immunizations (ORIs), which vaccinate populations in locations experiencing ongoing transmission, avoid these uncertainties. In practice, more than 75% of over 200 million doses deployed by 2024 were used for ORIs rather than preventive campaigns.[27]

Despite this widespread use, evidence on ORI impact remains fragmented. Most previous analyses have focused a limited number of large outbreaks,[2,5–7,12,14,25] which do not represent the diverse epidemiological profiles of cholera outbreaks,[25,32] or have overlooked key operational and immunological factors such ss delayed campaign initiation[27,33] or time required to develop vaccine-derived immunity.[13,14,17,18] A comprehensive modeling framework that integrates these factors an captures the epidemiological diversity of real-world outbreaks is therefore essential to guide evidence-based vaccine policy.

We addressed this evidence gaps by evaluating the health and economic impact of single-dose OCV ORIs across 1,192 cholera outbreaks in sub-Saharan Africa between 2010 and 2020. Specifically, we asked: (1) how effective are single-dose OCV ORIs when implemented under varying delays and coverage levels during individual outbreaks? and (2) how effective are ORIs when deployed across multiple heterogeneous outbreaks, as in global stockpile operations? To answer these questions, we combined outbreak-level impact estimates with aggregate analyses simulating vaccine deployment under stockpile constraints across a range of operational and prioritization scenarios.

Methods

Outbreak data

We compiled weekly suspected cholera incidence data from the Cholera Taxonomy Data,[34] which includes 1,664 outbreaks reported across various administrative levels in 30 countries in sub-Saharan Africa from January 2010 to July 2023.[25,35] Outbreaks were defined by location-specific average weekly incidence thresholds, spanning from the first to the last reported suspected case, with weeks without data assumed to have zero cases.[32]

To ensure consistency with real-world outbreak response conditions, several exclusion criteria were applied. Outbreaks were excluded if they occurred in administrative units whose population sizes fell outside the range of historical OCV deployments (i.e., between 16,500 and 5 million doses).[36] When outbreaks overlapped across multiple administrative levels, smaller outbreaks contained within a larger one were excluded. If a smaller administrative unit reported an equal or greater number of cases, the outbreak from the larger administrative unit was excluded. Outbreaks that occurred within three years of a prior OCV campaign were excluded to avoid vaccine-influenced incidence patterns that could bias counterfactual analyses. Finally, outbreaks beginning after Jan 1, 2021 were excluded due to incomplete OCV campaign records. After applying these criteria, 1,192 outbreaks were retained for modeling analyses. (**Figure 1A**).

Modeling the vaccine impact

We simulated weekly case counts under counterfactual vaccination scenarios—identical in all respects except for the implementation of ORIs. The impact of ORIs was based on the number of cases averted through vaccination (**Figure 1B**, top panel). Because ORIs can only occur if outbreaks are still ongoing, we assumed no campaign would be initiated if two or more weeks had passed since the outbreak ended, consistent with the outbreak definition[37] and validated in sensitivity analyses. ORIs beginning before outbreak resolution may still produce no impact due to the time required for implementation and vaccine-derived immunity to develop (**Figure 1B**, second panel). The impact of ORIs tends to be larger for a larger outbreak (**Figure 1B**, third panel**)** and the overall impact across multiple heterogeneous outbreaks tends to be larger than the simple average, as larger outbreaks contribute disproportionately to the overall impact (**Figure 1B**, fourth panel).

The model captures both direct and indirect effects of OCV[38] (Supplementary Information **S1**) and incorporates age-specific vaccine efficacy (< 5 and 5+ yo), delays in ORI implementation, and the delays in the vaccine-derived immunity (**Table 1**). The proportion of cases averted, $P_{i,\pi}$, at a given vaccine coverage π for each age group—children under five ($i = 1$) and individuals aged five or older ($i = 2$)—was computed as:

$$P_{i,\pi} = 1 - [\pi(1 - \psi_i)(1 - \eta_\pi) + (1 - \pi)(1 - \eta_\pi)],$$

where ψ_i represents direct vaccine efficacy for age group i and η_π denotes the indirect vaccine efficacy at coverage π. Thus, unvaccinated individuals, comprising $1 - \pi$ of the population, still benefit from indirect protection $1 - \eta_\pi$. We estimated η_π from reduction in incidence rates among unvaccinated individuals observed in a randomized trial conducted in Matlab, Bangladesh using Bayesian beta regression.[1,39] (Supplementary Information **S2** and **Figure S1**). The cumulative number of cases, $C_{i,\pi,\tau}$, for age group i, for a vaccination campaign initiated at time τ with coverage π was calculated as:

$$C_{i,\pi,\tau} = \sum_{t=1}^{T} I_{t,i}\left[1 - \mathbf{1}(t \geq \tau + \delta)P_{i,\pi}\right],$$

where $I_{t,i,}$ is the incidence at time t for age group i and $\mathbf{1}(x)$ is an indicator function (1 if x is true, 0 otherwise), δ captures delays for ORI implementation and the vaccine-derived immunity, and T marks the end of the outbreak. Overall, we simulated ORIs in each of the

1,192 outbreaks under combinations of vaccination coverage, campaign timing, and trigger criteria (outbreak onset-triggered or case-triggered) (**Table 2**). Two hundred parameter sets were drawn using a Sobol low-discrepancy sequence (**Table 3** and Supplementary Information **S3**) and results are summarized as medians with interquartile ranges (IQRs). The outcomes were reported as *impact*, i.e., cases, deaths, and disability-adjusted life years (DALYs) averted, which were also converted to *efficiency* (i.e., impact per 1,000 OCVs) and *cost-effectiveness* metrics (e.g., cost per unit of impact). All code and processed datasets are publicly available in a GitHub repository.[40]

To estimate the global impact of the ORI, we aggregated results across outbreaks assuming a constrained OCV supply of approximately 100,000 to 200 million doses, roughly reflecting the scale of global stockpile deployments.[36] Because this supply was insufficient to vaccinate all outbreak populations, we modeled allocation strategies across all or selected outbreaks prioritized by characteristics such as attack rate, duration, size, or case-fatality ratio. Outbreaks were sampled without replacement and assigned enough vaccine to reach their target coverage. Sampling continued until the cumulative number of doses matched the supply. This allocation was repeated 200 iterations.

The model captures on the immediate, within-outbreak impact of ORI and does not account for potential long-term benefits, such as preventing or mitigating future outbreaks. This assumption reflects the temporary population-level protection expected from single-dose campaigns, given waning immunity and population mobility.[9] Moreover, we modeled each population independently, without cross-population transmission or vaccination spillover.

Table 1. Model parameters.

Description	Values	Source
Parameters varied		

Direct vaccine efficacy by one dose of OCV for the < 5 yo and 5+ yo, respectively.	30% [95% CI: 15%, 42%] (< 5 yo) and 64% [95% CI: 58%, 70%] (5+ yo)	[30]
Campaign duration	5 [range: 3-14] days	[33]
Indirect vaccine efficacy	77.9% [95% CrI: 54.5 – 91] and 95% [95% CrI: 58.4 – 99.6] at 60% and 90% vaccine coverages, respectively. (Supplementary Information **S3** and **Figure S1**)	
Delay in vaccine-derived immunity	10.5 [range: 3, 21] days (Supplementary Information **S4**)	[41–49]
Parameters fixed		
Vaccine coverage	60%, 75%, and 90%	
Proportion of moderate and severe cholera	0.78, 0.22	[50]
Disability weight for moderate and severe diarrhea	0.188, 0.247	[51]
Vaccine cost of procurement	US$ 1.65 per dose	[52]
Vaccine delivery cost	US$ 2.32 per dose	[53]
Vaccine shipping cost	US$ 0.06 per dose	[54]
Cost to patient for cholera hospitalization and outpatient services	US$ 109.32 and US$ 16.06 per patient	[55]
Cost to public system for cholera hospitalization and outpatient services	US$ 184.97 and US$ 7.99 per patient	[55]
Number of workdays lost to patients and caregivers	6.5 and 3.3 days (Supplementary Information **S5**)	[56–59]
Mean age of infection	25 years (Supplementary Information **S6**)	[60–62]
Duration of illness	2 days	[63]
Life expectancy at a given age	Varies by year and country; 42.7 [IQR: 42.2 – 43.2] year of life	[64]

	expectancy for 25-year-olds during 2010-2020.	
Age distribution	Varies by year and country; 16.4% [IQR: 16.0, 16.9] of the population is under five during 2010 - 2020.	[64]
Gross Domestic Product (GDP)	Varies by year and country; 1 665 [IQR: 1 621 – 1 706] US$ during 2010-2020	[65]
Population in workforce	Varies by year and country; around 68.1% [IQR: 67.9 - 68.4] in during 2010-2020	[66]

Campaign triggers and assumptions

ORIs were triggered at the outbreak onset (outbreak onset-triggered) or when the cumulative number of reported cases exceeded a predefined threshold (case-triggered) (**Table 2**). The outbreak onset-triggered approach isolates the effect of response delay, clarifying how timing influences ORI impact. The case-triggered approach reflects potentially more realistic conditions in which uncertainty in surveillance or confirmation delays may postpone the start of campaigns. Under this scenario, identical case thresholds may trigger earlier responses in rapidly growing outbreaks and later responses in slower ones.

Vaccine coverage was varied across 60 %, 75 %, and 90 % of the affected administrative unit's population, consistent with ranges observed in global OCV campaigns (2013-2018)[67] and a recent campaign in Ethiopia.[68] We selected 75% as the representative single-dose coverage, reflecting levels typically achieved in humanitarian crises or outbreaks (Supplementary Information **S7**). The delay between vaccination and the onset of protection was based on observed seroconversion times (proxy for immune response),[41–49] ranging from 3 to 28 days (most often 7 to 14 days) after the first dose of a two-dose regimen[41,43–46] (Supplementary Information **S4**).

Table 2. Model settings for ORI (single-dose OCV campaign) settings

OCV campaign feature	Model settings that were tested
Campaign coverage	Target population included the entire population of the affected administrative unit. Three coverage levels were modeled: 60%, 75%, and 90%.
Campaign triggers	(i) Outbreak onset–triggered: ORI initiated in 1-15 weeks after outbreak onset, in 1-week increments; implementation set at the midpoint of the specified week. (ii) Case-triggered: ORI initiated once cumulative cases reached a predefined threshold (1, 5, 10, 30, 50, 70, 90, 140, 190, 240, 290 cases), followed by an additional 1–15 week implementation delay.
Vaccination rounds	Single-round vaccination campaign per outbreak
Outbreak data	Analyses performed at both the individual-outbreak level and in aggregated subsets selected by outbreak characteristics, including attack rate, duration, size, and case-fatality ratio.

Cost-effectiveness analysis

Cost-effectiveness was assessed as the cost per case, death, and DALY averted. Cost estimates were derived from published literature (**Table 1**; Supplementary Material **S8**). We adopted a societal perspective on cost, incorporating both direct costs (e.g., vaccine procurement and healthcare utilization) and indirect costs (e.g., productivity losses). All costs were converted to 2023 US dollars, adjusting for inflation using annual consumer price index changes for sub-Saharan Africa through 2023.[69–71]

Results

Outbreak data

We identified 1,192 cholera outbreaks across sub-Saharan Africa between 2010 and 2020. Cholera incidence displayed marked spatial and temporal heterogeneity at both regional and country levels. Monthly time series indicated that most outbreaks occurred in Eastern, Western, and Central Africa (**Figure 2A**). Outbreaks during 2010–2015 were concentrated in Western Africa, whereas those during 2016–2021 were more frequent in Eastern Africa. Central Africa experienced outbreaks more consistently throughout the period. Spatial heterogeneity was also evident at finer geographic scales (**Figure 2B**). Outbreaks typically peaked around week 5 (mean = week 6; median = week 4), and approximately 88% peaked before week 10 (**Figure 2C**). Outbreak durations ranged from 5 to 93 weeks, with sizes from 3 to 14,522 suspected cases and the attack rates from 0·005 to 112 rates per 10,000 populations (**Figures 2D-E**). Among the 152 outbreaks reporting at least one laboratory-confirmed case, duration ranged from 5 to 56 weeks, outbreak sizes from 10 to 7,726 suspected cases, and attack rates from 0.005 to 19 rates per 10,000 population. Outbreaks excluded from analysis exhibited similar spatiotemporal patterns, although they tended to be larger. These consistencies suggest that the dataset is broadly representative of cholera epidemiology across the region (**Figure S2**).

Impact of ORIs in individual outbreaks

Impact of ORIs across campaign timing and coverage levels

In the outbreak-onset-triggered scenario, an ORI launched in week 1 at 75% coverage averted a median of 70% [IQR: 49.8-82.5] of cases—corresponding to 129 cases [IQR: 41-375] averted (**Figure 3A**). Higher coverage yielded greater benefit: at week 1, the median cases averted were 120, 129, and 135 for coverage levels of 60%, 75%, and 90%, respectively. By week 9, these values fell to 15, 16, and 17 cases, corresponding to < 9% of cases across all coverage levels. Substantial variability remained for each delay, reflecting heterogeneity in outbreak characteristics and campaign parameters.

An exponential decay model captured this decline (R^2= 0.90 [IQR: 0.78 - 0.96]) (**Figure 3B**). Among outbreaks lasting >=7 weeks (91% of outbreaks), for which ORIs could be implemented as late as week 9, the decay rate was 0.27 [IQR: 0.11 - 1.49] per week, corresponding to a weekly decline of 23.6% [IQR: 10.7 - 77.4] % and a halving time of 2.6

[IQR: 0.5 - 6.1] weeks. The decay was slower in longer outbreaks (weekly decline of 19.2 [IQR: 9 - 61.7] for outbreaks lasting >= 17 weeks; 32% of outbreaks). Delay also caused some outbreaks to end because vaccination began (**Figure 3C**). From week 5 onward, an additional 5.4% of outbreaks were missed per week of delay, reaching around 25.3% by week 9.

Efficiency of ORIs across campaign timing and prioritization

Prioritizing larger outbreaks produced the greatest absolute case reduction (**Figure 3D; Figure S3** in the Supplementary Information) whereas prioritizing high-attack-rate outbreaks maximized ORI efficiency (**Figures 3E–F**). Targeting the top 10% of outbreaks by size increased the median number of cases averted to 682 cases (vs. 16 under random targeting) for week-9 ORIs, and to 1,725 cases for week-1 ORIs. Prioritizing the top 10 % by attack rate increased the ORI efficiency approximately to 9.58 cases (1.28 DALYs) and 2.39 (or 0.06 DALYs) averted per 1,000 doses for week-1 and week-9 ORIs, respectively. Prioritizing by size or duration also improved efficiency but to a lesser extent (**Figure S3**). Percentage cases averted, which was not normalized by doses, was most sensitive to outbreak duration. The median percentage cases averted were 83.8%, 82%, and 80.1% for outbreaks in the top decile of duration, size, and attack rate, respectively, for week-1 ORIs, declining to 55.2%, 26.7%, and 17.1% by week 9.

Table 3 summarizes the impact, efficiency, and cost effectiveness of the outbreak onset-triggered ORI with 75% vaccine coverage (results for other coverage levels and delays appear in **Table S1**). As delays increased, the impact and the efficiency of ORIs declined, while costs per unit of impact rose. The cost per case averted increased from 4,662 [IQR: 985 - 17,917] for week-1 ORIs to US$13,043 [IQR: 2,575 - 60,026] for week-9 ORIs. Corresponding costs per DALY averted rose from US$24,770 [IQR: 4,066 - 4,679,419] to US$1,745,455 [IQR: 11,816 - 40,406,869], exceeding the median GDP per capita in sub-Saharan Africa ($1,665[65]) in 2010 - 2020. Impact, efficiency, and cost effectiveness were consistently greater among outbreaks with higher attack rates, longer duration, or larger size. In the top quartile by duration, percentage case averted reached 82% [IQR: 72.9 - 88] and 41.3% [IQR: 25.6 - 57.4] for ORI launched in week 1 and week 9, respectively. Targeting the top quartile by attack rate yielded the most favorable efficiency and cost effectiveness outcomes. Week-1 and week-9 ORIs cost US$ 2,967 [IQR: -35 - 46,131] and US$ 20,433 [IQR: 3,474 - 2,777,567] per DLAY averted, respectively—approximately 1.8 and 12.3 times the GDP per capita.

Case-triggered scenario

For the case-triggered scenario, the impact, efficiency, and cost effectiveness declined with increasing implementation delays and higher case thresholds triggering vaccination (**Table 4**). Median percentage case averted for week-1 ORIs ranged from 70%, as in the outbreak-onset triggered scenario, to below 35% in case at least 140 cases were required to trigger a response. An exponential decay model (R^2 = 0.87 [IQR: 0.77 - 0.93]) predicted that case averted decreased by 6.4% [IQR: 2.5 - 14.2] for every 10 additional cases missed before response initiation, corresponding to halving of impact every 104.7 additional cases [IQR: 45.3 - 272.7].

ORIs across multiple outbreaks

Impact across campaign timing, OCV doses, and prioritization

Assuming 100,000 to 200 million OCV doses, the number of outbreaks eligible for ORI ranged from 1 to 559 when random targeting. Because outbreaks varied in population size, prioritizing larger outbreaks reduced the number targeted (435 [IQR: 420-453] outbreaks for 200 million doses), whereas prioritizing those with higher attack rate increased it (776 [IQR: 769-785]), reflecting the tendency of high-attack-rate outbreaks to be smaller (**Figure 4A**). Total cases averted increased with both the number of doses deployed and prioritization (**Figure 4B**). For ORIs implemented in week 9, the median number of cases averted per 200 million doses was 72,362 [IQR: 66,407 - 79,326] under random targeting, compared with 89,347 [IQR: 83,640 - 95,170], 96,929 [IQR: 91,991 - 104,011], and 156,732 [IQR: 154,407 - 158,981] when prioritizing by outbreak size, duration, and attack rate, respectively. When the delay was shortened to 1 week, the corresponding medians increased to 200,283, 247,062, 260,528, and 422,804, respectively.

Efficiency and cost effectiveness across campaign timing, OCV doses, and prioritization

ORI efficiency varied widely at low vaccine supply (<= 200,000 doses) but converged as supply increased (**Figure 4C**). Targeting high-attack-rate outbreaks consistently achieved the greatest efficiency, exceeding 2 cases averted per 1,000 doses for week-1 ORIs and a bit lower than 1 case per 1,000 doses for week-9 ORIs, while other targeting strategies remained lower. The pattern for DALYs averted per 1,000 doses mirrored that of cases averted per 1,000 doses, though with smaller magnitude—approximately 0.6 for high-attack-rate targeting and 0.3–0.4 for other targeting strategies for week 1 ORIs (**Figure 4D**).

Cost effectiveness improved both with both shorter delays and targeted deployment (**Figures 4E-F**). The cost per case averted decreased from US$ 8,352 [IQR: 7,641-9,140] under random targeting to US$ 4,298 [IQR: 4,237-4,368] when high-attack-rate outbreaks were prioritized with week-9 ORIs, and further to US$ 3,410 [IQR: 3,173-3,615] and US$ 1,340 [IQR: 1,321-1,362] for random and high-attack-rate targeting, respectively, with week-1 ORIs (**Figure 4E**). Similarly, cost per DALY averted declined from US$ 31,157 [IQR: 28,533-34,696] to US$ 16,465 [IQR: 16,205-16,752] for week-9 ORIs, and from US$ 11,531 [IQR: 10,919 - 12,479] to US$ 4,688 [IQR: 4,624 - 4,786] for week-1 ORIs (**Figure 4F**).

Discussion

Our study demonstrated the substantial potential impact of OCVs to reduce the burden of cholera outbreaks, based on modeling of 1,192 outbreaks in sub-Saharan Africa from 2010 to 2020. We found that the timing of ORI was critical: the median percentage of cases averted declined from 70.0% to 7.8% when implementation was delayed from week 1 to week 9. This corresponds to a roughly 23.6% reduction in impact per week of delay, with the benefit halving every 2.6 weeks. ORI efficiency showed a similar decline, falling from roughly 0.69 to 0.09 cases averted per 1,000 doses (or vaccinated persons) over the same delay interval, corresponding to $US 4,662 and $US 13,043 per case averted, respectively.

The impact of ORI varied not only by campaign parameters (coverage and delay) but also by outbreak characteristics, which differ widely across settings in sub-Saharan Africa. Consequently, uncertainty intervals for percentage cases averted were broad, often ranging from near 0% to 100%. Delays in ORI implementation could be partially offset by prioritizing outbreaks that were larger, longer lasting, or had higher attack rates in the absence of vaccination. For example, a median 22.8% case reduction for week-5 ORIs across all outbreaks(was comparable to 24.6% for week-13 ORIs in the top quartile by duration. Similarly, median ORI efficiency increased from 0.20 cases averted per 1,000 doses from an week-5 ORIs across all outbreaks to 0.55 when targeting outbreaks in the top quartile by attack rate from an week-13 ORI. These findings, consistent with previous analyses highlighting the benefits of selective targeting in sub-Saharan Africa,[5,72] underscore the value of strategic prioritization. Allocating limited OCV supplies to outbreaks most likely to become extensive or prolonged could markedly increase the impact, efficiency, and cost effectiveness of ORIs. Although such prioritization assumes an ability to predict outbreak trajectory at the time of response, this may be increasingly feasible using environmental and socioeconomic indicators, and WASH conditions, and historical outbreak trends.[21,73–76] As predictive models and surveillance systems improve, they can guide more targeted and efficient reactive vaccination.

Our analysis of case-triggered ORIs further underscores the importance of early detection as well as rapid implementation. For week-1 ORIs, the median percentage of cases averted declined from 70% when outbreak response was triggered at outbreak onset to below 35% when triggered only after more than 100 cases had already occurred—equivalent to surveillance missing the first 100 cases or these cases going undetected. These findings highlight the need for robust surveillance systems capable of rapidly detecting and confirming cholera cases, alongside operational and logistical capacity to deliver vaccines and initiate campaigns immediately once response thresholds are reached.

At the aggregate level, the overall impact of ORI did not scale linearly with the median outbreak-level estimates. When successive ORIs were implemented across multiple outbreaks, large and long-duration outbreaks—which yielded greater benefits from ORIs—contributed disproportionately to the overall benefit, moving the overall median upward. Assuming 200 million OCV doses (comparable to the total deployed globally since 2013), our model estimated that random targeting would avert around 0.45 cases per 1,000 doses (compared to 0.1 at the outbreak level), increasing to 2.13 when prioritizing high-attack-rate outbreaks, equivalent to about 422,000 cases averted for week-1 ORIs.

Our estimates of impact, efficiency, and cost-effectiveness are broadly consistent with, but likely more representative than, previous studies that focused on a few large-scale outbreaks.[2,3,5,7,8,10,12,15,19,77]. For instance, analyses based on a few outbreaks have reported roughly 30–90% case reductions from reactive campaigns in Bangladesh, Zimbabwe, Guinea, Haiti, Chad, and Thailand,[1,5,13,14] often assuming high coverage or early response. By incorporating over a thousand outbreaks of varying epidemiologic characteristics, our approach provides regionally generalizable estimates across realistic operational settings. The ORI efficiency in our study (2.13 cases averted per 1,000 doses when prioritizing high-attack-rate outbreaks) aligns with the estimates from Xu et al.,[78] who reported 2.6-11 cases averted per 1,000 fully vaccinated persons (two doses) in preventive campaigns across sub-Saharan Africa. Although higher efficiencies have been reported from campaigns in Bangladesh,[79], Indonesia, Nigeria, and Uganda[80], and Guinea, Haiti, Zimbabwe, Haiti,[14] these likely reflect conditions not representative of most outbreaks. Likewise, our estimated cost per DALY averted (~US $ 24,770 and US $ 2,977 for random targeting and the top quartile by attack rate with week-1 ORIs) falls within previously reported ranges (US $1,800–21,000 per DALY averted) for two-dose preventive campaigns in sub-Saharan Africa.[72]

This study has several limitations. First, we assumed uniform vaccine coverage within administrative units, which may underestimate the efficiency of spatially focused strategies such as case-area targeted interventions[5,81–83] or district-level geographic targeting.[72,78] Nonetheless, our findings showing higher efficiency of ORIs in outbreaks with high attack rates suggest that these conditions can approximate the benefits of spatially targeted approaches. Conversely, unaccounted vaccine wastage[84] could lead to overestimation of ORI efficiency and cost effectiveness. Second, our parameterization of indirect protection relied on data from Bangladesh,[1,39] which may not fully generalize to sub-Saharan African contexts; nevertheless, the consistency between our estimates and the outcomes from the dynamic model (Supplementary Information **S9; Figures S5-S7**) suggests that modeled indirect effects are robust. Third, we modeled each outbreak independently and thus the impact of ORI was confined to the specific outbreak in which the ORI was implemented. In reality, ORIs might reduce future outbreaks in the same or neighboring areas given that two-dose OCVs provide protection for up to 4-5 years.[85–88] Due to a lack of geographic details for each outbreak and epidemiologic links between the outbreaks, we could not evaluate the downstream or spatial spillover effects. Fourth, as roughly half of reported suspected cholera cases are not true *Vibrio cholerae* infections.[89] our reliance on suspected case data may overestimate impact; however, underreporting of cholera in sub-Saharan Africa may offset this bias[90]. Fifth, we assumed ORIs would not be implemented if more than two weeks had elapsed after outbreak resolution, whereas in practice, reactive campaigns may still occur. Sensitivity analyses confirmed a monotonic decline in ORI impact and efficiency as the delay to stop ORI after outbreak end increased (**Table S2**). Sixth, our outbreak dataset was generated using standardized but operational definitions that vary by context. Alternative outbreak criteria could yield different datasets and, consequently, different estimates. The main analysis excluded outbreaks from administrative units with populations that were outside of typical range of OCV deployments (16,500-5,000,000). To explore a broader range of outbreak settings, we relaxed these criteria and included outbreaks occurring in populations between 5,000 and 10,000,000. Under this more inclusive scenario, the median and interquartile ranges remained largely unchanged (**Table S3**). Finally, our simulation of OCV deployment from the global stockpile simplified several realistic details—such as characteristics of outbreaks selected, variability in coverage, and the full distribution of implementation delays—to highlight the disproportionate impact of long and large outbreaks to overall ORI impact and the benefits of prioritization by outbreak characteristics. These results should therefore be interpreted qualitatively.

Conclusion

Our study quantifies the health and economic benefits of timely and strategically targeted OCV outbreak response across diverse epidemiological settings in sub-Saharan Africa. Rapid ORI implementation and prioritization of large, prolonged, or high–attack-rate outbreaks can maximize health gains and cost-effectiveness, even under vaccine supply constraints. Strengthening surveillance, improving predictive tools, and ensuring rapid vaccine deployment will be essential to realize the full potential of reactive cholera vaccination.

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Author contributions

J-HK and ECL conceptualized the study. J-HK led the study design and analysis. J-HK and MD performed the statistical analysis and drafted the initial manuscript. ECL and QZ collected and prepared the data for analysis. ECL, ASA, and JHK provided critical input on methodology. MD conducted the literature review. All authors participated in the interpretation of the findings, critically reviewed the manuscript for intellectual content, and approved the final version for submission.

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